Primitive recursion preserves lambda definability

The Open Logic Project contributors

openlogic:lambda-calculus/lambda-definability:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned source file sources/olp-lambda-calculus-lambda-definability/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Card cut from the FREE slice of the lit pool for unit olp-lambda-calculus-lambda-definability; the proof text reuses Succ, pairing, and projections from earlier sections of the same unit without cross-unit references, so no navigation stubs are needed. The declared simplification to one additional argument is recorded in the difficulty reason.

Editorial scope. Single lemma card cut from the Open Logic Project unit olp-lambda-calculus-lambda-definability (pinned-original-source.tex); statement and proof are the author's own text at source lines 1145-1150 and 1152-1226.

Allowed standard prerequisites (editorial).

- Pairing and projections (lambda-definability unit, earlier section: terms for ordered pairs with first and second projections). used at proof lines 1166-1170 and 1172-1177. The iteration state is a pair (y, h-value) maintained by the update term D and read out by the second projection.
- Church numerals as iterators (lambda-definability unit, earlier section). used at proof lines 1179-1182 and 1212-1224. The numeral for m applies its argument D exactly m times, so H at numerals n and m reduces to the second projection of the m-fold iterate of D at the initial pair.
- Successor term (lambda-definability unit, Proposition prop:succ-ld section, the term Succ). used at proof lines 1169-1170 and 1204-1207. The update term D refreshes the first component of the state pair with the successor.

Lemma 0.1. *Suppose f is an n-ary function and g is an $n+2$-ary function, they are lambda definable by terms F and G, and the function h is defined from f and g by primitive recursion. Then h is also lambda definable.*

Proof. Recall that h is defined by

$$\begin{aligned} h(x_1, \ldots, x_n, 0) &= f(x_1, \ldots, x_n) \\ h(x_1, \ldots, x_n, y+1) &= h(x_1, \ldots, x_n, y, h(x_1, \ldots, x_n, y)). \end{aligned}$$

Informally speaking, the primitive recursive definition iterates the application of the function h y times and applies it to $f(x_1, \ldots, x_n)$. This is reminiscent of the definition of Church numerals, which is also defined as a iterator.

For simplicity, we give the definition and proof for a single additional argument x. The function h is lambda defined by:

$$H \equiv \lambda x.\, \lambda y.\, \text{Snd}(yD\langle \overline{0}, Fx\rangle)$$

where

$$D \equiv \lambda p.\, \langle \text{Succ}(\text{Fst } p), (Gx(\text{Fst } p)(\text{Snd } p))\rangle$$

The iteration state we maintain is a pair, the first of which is the current y and the second is the corresponding value of h. For every step of iteration we create a pair of new values of y and h; after the iteration is done we return the second part of the pair and that's the final h value. We now prove this is indeed a representation of primitive recursion.

We want to prove that for any n and m, $H\,\overline{n}\,\overline{m} \twoheadrightarrow \overline{h(n,m)}$. To do this we first show that if $D_n \equiv D[\overline{n}/x]$, then $D_n^m\langle \overline{0}, F\,\overline{n}\rangle \twoheadrightarrow \langle \overline{m}, \overline{h(n,m)}\rangle$ We proceed by induction on m.

If $m = 0$, we want $D_n^0\langle \overline{0}, F\,\overline{n}\rangle \twoheadrightarrow \langle \overline{0}, \overline{h(n,0)}\rangle$. But $D_n^0\langle \overline{0}, F\,\overline{n}\rangle$ just is $\langle \overline{0}, F\,\overline{n}\rangle$. Since F lambda defines f, this reduces to $\langle \overline{0}, \overline{f(n)}\rangle$, and since $f(n) = h(n,0)$, this is $\langle \overline{0}, \overline{h(n,0)}\rangle$

Now suppose that $D_n^m\langle \overline{0}, F\,\overline{n}\rangle \twoheadrightarrow \langle \overline{m}, \overline{h(n,m)}\rangle$. We want to show that $D_n^{m+1}\langle \overline{0}, F\,\overline{n}\rangle \twoheadrightarrow \langle \overline{m+1}, \overline{h(n,m+1)}\rangle$.

$$\begin{aligned} D_n^{m+1}\langle \overline{0}, F\,\overline{n}\rangle &\equiv D_n(D_n^m\langle \overline{0}, F\,\overline{n}\rangle) \\ &\twoheadrightarrow D_n\,\langle \overline{m}, \overline{h(n,m)}\rangle \qquad \text{(by IH)} \\ &\equiv (\lambda p.\, \langle \text{Succ}(\text{Fst } p), (G\,\overline{n}(\text{Fst } p)(\text{Snd } p))\rangle)\langle \overline{m}, \overline{h(n,m)}\rangle \\ &\to \langle \text{Succ}(\text{Fst } \langle \overline{m}, \overline{h(n,m)}\rangle), \\ &\qquad (G\,\overline{n}(\text{Fst } \langle \overline{m}, \overline{h(n,m)}\rangle)(\text{Snd } \langle \overline{m}, \overline{h(n,m)}\rangle))\rangle \\ &\twoheadrightarrow \langle \text{Succ } \overline{m}, (G\,\overline{n}\,\overline{m}\,\overline{h(n,m)})\rangle \\ &\twoheadrightarrow \langle \overline{m+1}, \overline{g(n,m,h(n,m))}\rangle \end{aligned}$$

Since $g(n, m, h(n,m)) = h(n, m+1)$, we are done.

Finally, consider

$$
\begin{aligned}
H\,\overline{n}\,\overline{m} &\equiv \lambda x.\,\lambda y.\ \mathrm{Snd}(y(\lambda p.\langle \mathrm{Succ}(\mathrm{Fst}\ p), (G\,x\,(\mathrm{Fst}\ p)\,(\mathrm{Snd}\ p))\rangle)\langle \overline{0}, Fx\rangle) \\
&\quad\ \overline{n}\,\overline{m} \\
&\twoheadrightarrow \mathrm{Snd}(\overline{m}\ \underbrace{(\lambda p.\langle \mathrm{Succ}(\mathrm{Fst}\ p), (G\,\overline{n}\,(\mathrm{Fst}\ p)(\mathrm{Snd}\ p))\rangle)}_{D_n}\langle \overline{0}, F\overline{n}\rangle) \\
&\equiv \mathrm{Snd}(\overline{m}\,D_n\,\langle \overline{0}, F\overline{n}\rangle) \\
&\twoheadrightarrow \mathrm{Snd}\,(D_n^m\langle \overline{0}, F\overline{n}\rangle) \\
&\twoheadrightarrow \mathrm{Snd}\,\langle \overline{m}, \overline{h(n,m)}\rangle \\
&\twoheadrightarrow \overline{h(n,m)}.
\end{aligned}
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